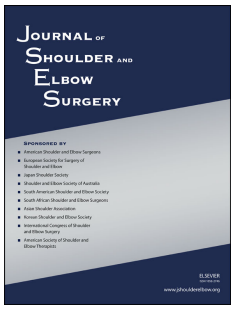
Journal Pre-proof



The Increasing Medicolegal Cost in the Care of High-Level Athletes

Luke D. Latario, MD, Carly J. Deter, B.S., Abby M. Deter, J. D., Mark E. Baratz, MD

PII: S1058-2746(24)00581-0

DOI: https://doi.org/10.1016/j.jse.2024.07.011

Reference: YMSE 6983

To appear in: *Journal of Shoulder and Elbow Surgery*

Received Date: 9 March 2024

Revised Date: 29 June 2024

Accepted Date: 4 July 2024

Please cite this article as: Latario LD, Deter CJ, Deter AM, Baratz ME, The Increasing Medicolegal Cost in the Care of High-Level Athletes, *Journal of Shoulder and Elbow Surgery* (2024), doi: https://doi.org/10.1016/j.jse.2024.07.011.

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2024 Published by Elsevier Inc. on behalf of Journal of Shoulder and Elbow Surgery Board of Trustees.

Journal Pre-proof

The Increasing Medicolegal Cost in the Care of High-Level Athletes

Running Title: Increasing Cost in Caring for Athletes

Luke D. Latario MD, Carly J. Deter B.S.,† Abby M. Deter J. D.‡, Mark E. Baratz MD*

*University of Pittsburgh Medical Center Department of Orthopaedic Surgery, Pittsburgh, PA, USA

†Geisinger Commonwealth School of Medicine, Scranton, PA, USA

‡University of Pittsburgh School of Law, Pittsburgh, PA, USA

Corresponding author: Luke D. Latario MD luke.latario@gmail.com

Geisinger Commonwealth School of Medicine, 525 Pine St, Scranton, PA 18509, USA

Disclaimers:

Funding: No funding was disclosed by the authors.

Conflicts of interest:

The authors, their immediate families, and any research foundation with which they are affiliated have not received any financial payments or other benefits from any commercial entity related to the subject of this article.

Institutional Review Board approval was not required for this retrospective project using a legal research database without protected health information.

Journal Pre-proof

Journal Pre-proof

Abstract

Background:

Medical malpractice represents a significant economic cost in healthcare. Increasingly large damage claims by professional athletes against physicians have raised concerns about the medicolegal challenges in caring for high-level athletes.

Methods:

An online proprietary legal research database was queried for lawsuits related to malpractice in the care of professional and amateur athletes from 1992-2023. Demographics of plaintiffs and defendants, details of lawsuit filings and damages claimed were recorded for all cases, settlements, and jury verdicts. Descriptive statistics, linear regression, as well as univariate analysis of demographic factors and damage claims in inflation-adjusted 2023 dollars was performed.

Results:

Eighty-eight lawsuits were recorded from 1992-2023. The sum of indemnities exceeded 186 million in 2023 inflation-adjusted United States Dollars. Football players were the most commonly represented athletes (n=26) and represented 49% of total financial awards. Fourteen cases (16%) involved treatment of the upper extremity. Professional and collegiate level of play was associated with higher damages in favor of plaintiffs when compared to other levels of play. No other demographic was associated with higher financial awards. Linear regression showed a significant positive trend with an increasing inflation-adjusted compensation for plaintiff verdicts in the last 30 years.

Journal Pre-proof

Journal Pre-proof

Conclusions:

There is an increasing medicolegal financial risk associated with the care of athletes. This is higher in collegiate and professional levels of play. As physicians, insurers and institutions adjust to these financial risks, care must be taken to avoid ramifications on the availability and quality of care provided to athletes. Shoulder and elbow surgeons may consider additional preoperative counseling, legal waiver forms regarding malpractice claims and advocacy for medical malpractice reforms in the care of athletes with high earning potential to mitigate these increasing financial risks.

Level of Evidence: Level IV; Descriptive Epidemiology Study

Malpractice litigation presents a high economic burden on the United States healthcare system. There is an estimated 4 billion dollars annually in direct cost associated with malpractice indemnities and settlements,[17] and an estimated 60 billion dollar annual cost to the healthcare system as practitioners perform "defensive" medical care.[3, 16, 18, 20] These costs stress financially challenged hospitals and health systems across the United States.[2]

Surgeons and physicians caring for the athlete face unique challenges related to the higher expectations and functional demands of this patient population, and the numerous responsibilities of a team physician.[3] Shoulder and elbow surgeons face particularly high medicolegal risk with 15% of orthopedic surgeons facing a lawsuit every year.[3, 7, 9, 16, 17] An analysis of 26 years of publicly available orthopedic malpractice lawsuits 14% of all cases involved sports or shoulder/elbow pathology.[7] The most common cause for litigation involving orthopedic sports procedures are pain and suffering, the need for multiple procedures and neurovascular injury.[8] However, this risk is not isolated to orthopedic surgeons with other

Journal Pre-proof

Journal Pre-proof

physicians providing game coverage, clearance for sports or return to play facing similar medicolegal challenges and liability in caring for athletes.[13, 14]

Recent high profile settlements involving professional athletes generated debate and concern regarding the perilous professional and medical-legal environment for physicians caring for athletes. In February 2023, a former National Football League athlete was awarded 43.5 million dollars related to treatment of a PCL injury at the age of 30.[5] Another Former NFL player was awarded 28.5 million dollars after treatment for an ankle fracture sustained while playing on a four-year 2.3 million dollar contract with the New York Giants in 2014.[4] Similarly, there are national trends across all United States lawsuits where cases that resulted in verdicts exceeding 10 million dollars in indemnities, noneconomic damages exceed both punitive and economic damages combined in the majority of instances.[19]

In response to these and other recent high-profile lawsuits, an open letter calling for protection of team physicians and their unique interdisciplinary role in caring for athletes signed by 23 academic medical societies was published.[1, 11] Despite these concerns, there is limited research on the medical malpractice burden facing orthopedic surgeons in the treatment of athletes. The purpose of this study is to quantify and describe the trends in malpractice litigation involving the care of athletes at the professional and collegiate level using a publicly available database.

Materials and Methods:

Westlaw Precision is a database of United States legal cases, lawsuits and jury . It was queried from January 1st, 1993 to December 31st, 2023. Cases and Jury verdicts were searched using two keywords entries, the first being "Sport" and any of the following with the Boolean operator "OR" "Orthopedic, Surgeon, Physician, Malpractice, Athlete" and the second being

Journal Pre-proof

Journal Pre-proof

"Athlete" and any of the following with the Boolean operator "OR" "Orthopedic, Surgeon, Physician, Malpractice, Sport." Filters related to medical malpractice were then applied. Cases including care of athletes at all levels with damages related to their ability to participate in future recreation and sports activities were included. Obstetrical injury cases were excluded.

Demographic factors of defendants and plaintiffs were recorded as were financial rewards in cases in favor of the plaintiffs. Data was collected between February 3rd, 2023 and November 14th, 2023. Financial indemnities were converted to inflation adjusted 2023 United States Dollars (USD) using an online calculator provided by the Federal Reserve Bank of Minneapolis.[10]

Statistical analysis was performed using Graphpad prism v10.1.2 (GraphPad, San Diego, CA, USA). Univariate analysis was performed using Mann-Whitney U test for nonparametric data. A linear regression was used for the inflation-adjusted cost of damages for cases found in favor of plaintiffs and the year of the lawsuit.

Results:

Eighty-eight cases were identified in the search. The majority involved were male athletes (n=66) who were adult (n=41) and with a mean age of 23.1 years.(Table 1) Football was the most commonly represented sport (n=26). The most common level of play was high school athletes (n=30). Among the defendant subspecialties orthopedics was the most prevalent (n=40. (Table 1)

Thirty-seven cases were found in favor of the plaintiff with 28 jury verdicts, 6 settlements and 4 arbitrations. (Table 2) Forty-five jury verdicts were in favor of the defendants. Remaining case outcomes were not reported in the database or were reported as confidential. The sum of reported indemnities exceeded 186 million 2023 USD, and football players represented 49% of

Journal Pre-proof

these damages. College and professional athletes represented 16% and 42% of total financial indemnities respectively. (Table 2)

Univariate analysis of demographics found only higher levels of play associated with higher median damages awarded. (Table 3) Defendant specialty, plaintiff sex, participation in football, wrongful death, or adult or pediatric age was not found to be statistically significant. (Table 3) Linear regression analysis showed an increasing trend in the inflation-adjusted cost of financial awards to plaintiffs (p=0.014). (Figure 1)

Discussion:

Our study showed players at a professional or collegiate level involved in medical malpractice litigation commanded higher indemnities than other malpractice lawsuits involving athletes. No other plaintiff or defendant demographic in this study was associated with a difference in inflation-adjusted indemnities. There was also a significant increasing trend in the size of damages awarded over the last 30 years.

In this study the majority of cases were found in favor of the defendant, which is consistent with previous work published on malpractice lawsuits involving orthopedic surgeon malpractice[6, 8] and sports-related spine injuries.[22] Additionally, orthopedic surgeons and surgical specialties, were not associated with higher indemnities in cases decided in favor of plaintiffs when compared to other medical providers. This held true even though over half of the cases found in favor of plaintiffs were lawsuits against orthopedic surgeons. Previous work has shown orthopedic surgery is associated with a high burden of medical malpractice litigation,[9] with the majority of orthopedic surgeons being involved in at least one malpractice legal action over their career.[7, 9, 16, 17] Additionally, since 1988, the frequency of orthopedic malpractice legal action has increased over 200%.[7] Our study suggests that the increasing liability found in providing care for

Journal Pre-proof

Journal Pre-proof

113 high level athletes is not restricted to orthopedic surgeons, but may contribute to the already high
114 risk associated with orthopedic practice. Forty five percent of all lawsuits in our study involved
115 orthopedic physicians. (Table 1) There are an estimated 29,000 orthopedic surgeons practicing
116 in the United States[12] making up approximately 3% of the total 950,000 practicing physician
117 workforce in the United States. [21] This represents a 15-fold increase in exposure to litigation in
118 the care of athletes for orthopedic surgeons compared to other providers.

119 In the 37 cases with awards reported to the plaintiff, the total cost of settlements in the
120 care of athletes from 1992-2023 in 2023 inflation adjusted dollars was $186,612,214. This is a
121 relatively small sum compared to the 4 billion dollars annually in direct cost associated with
122 malpractice indemnities and settlements in the United States.[3, 16-18, 20] However the average
123 award was in excess of 5 million dollars and the largest was 43.5 million. By comparison, the
124 mean payout for orthopedic medical malpractice lawsuits across all subspecialties and patients in
125 the United States has been reported between $898,972 -$1,374,640.[7, 16] Settlements of this size
126 can create existential threats to independent practitioners or smaller institutions.[2, 11, 19] It is also
127 unclear how this affects resource utilization and medical decision-making in healthcare systems
128 participating the care of high level athletes and the physician-athlete relationship.

129 Shoulder and elbow surgeons should consider the impact of the care of athletes in their
130 malpractice liability. In a study of 25 medical malpractice lawsuits involving shoulder and
131 elbow surgery between 2010-2020 Sharma et al[18] found the mean payout to be $4,058,952. The
132 average indemnity for athletes in our study was higher at $5,314,867 in 2023 inflation adjusted
133 dollars for the 3 cases found in favor of the plaintiff. (Table 2) Unfortunately this study suggests
134 treatment of high level athletes carries increasing malpractice liability for surgeons.

Journal Pre-proof

Journal Pre-proof

As surgeons assume greater financial risk in the care of professional athletes, potentially injurious biases or hesitancies in treatment of athletes may be introduced. These concerns were highlighted in the open letter published in June of 2023 by the American Orthopaedic Society for Sports Medicine stating "litigation may have an enormous negative impact on the medical care of competitive and elite athletes."[1] In the foreseeable future physicians may electively choose to limit exposure to this increasing financial liability. This may occur by physicians willingly reducing involvement in care of professional athletes, or at the behest of their employing hospital and academic institutions seeking to reduce or avoid partnerships with teams to mitigate economic risk. An alternative strategy is institutions requesting that athletes sign waivers that limit awards or prescribe where a medical malpractice case can be tried. All of this may hamper the subspecialized and multidisciplinary care required for the treatment of injured high performing athletes. Concerned physicians may consider collaborating with administrative leadership regarding the creation of ethical policies and procedures around any waivers for the care of athletes designed to mitigate financial risk. However, any such waiver may not be enforceable and the authors are not aware of any related medical malpractice case law precedent within the United States. As modern high level athletes frequently compete internationally travelling the globe,[15] physicians and medical professional societies both in the United States and internationally should consider advocacy for additional legal protections or limitations on malpractice award. This would be designed to avoid "nuclear verdicts"[19] where a single lawsuit's exorbitant financial indemnity threatens closure or restructuring of medical practices and institutions.

Limitations of this study include those inherent to retrospective and database studies. Westlaw may not contain all lawsuits related to the malpractice and the care of athletes, and

Journal Pre-proof

Journal Pre-proof

entries may not be complete or contain keywords to be accurately included in this study. Additionally, lawsuits involving high performing athletes are relatively rare due to the low number of professional athletes nationwide and their likelihood of having an adverse outcome from care prompting a malpractice claim.

Conclusions

There is a trend towards increasingly large damage claims over the last 30 years in malpractice lawsuits involving athletes, particularly in the care of collegiate and professional level players. This is concerning for possible erosion of the close working relationship of many shoulder and elbow surgeons and other physicians with athletes and sports teams.[1, 11] Future work is needed to identify measures to increase communication, trust and patient satisfaction in athletes receiving care, and systems for financial protections for physicians facing inordinately large financial indemnities.

Table 1. Demographics, timeline, and sports participation in malpractice lawsuits involving care of athletes. Values in brackets [ ] denote percentages.

Table 2. Number of lawsuits and mean financial awards by level of play, and sport. Financial damages reported in 2023 inflation adjusted United States Dollars (USD).

Table 3. Univariate analysis of damages awarded by demographic factors in cases found in favor of plaintiff.

Figure 1. Financial damages awarded to plaintiffs in 2023 inflation-adjusted dollars by year. Linear regression analysis showed a positive slope with p-value of p=0.0140.

Figure 2. Financial damages awarded to plaintiff in 2023 inflation adjusted dollars by year between high level (professional and collegiate athletes) and all other athletes. Linear regression

Journal Pre-proof

Journal Pre-proof

was positive for both groups with a significant p-value of 0.0431 for high level athletes and 0.2659 for all others.

References

1. Protecting Athletes and Team Physicians Who Care for Them Amid Rising Complexity and Liability Risk. American Orthopaedic Society for Sports Medicine.
2. The Financial Stability of America's Hospitals and Health Systems Is at Risk as the Costs of Caring Continue to Rise. www.aha.org: American Hospital Association 2023:1-9.
3. Bal BS. An introduction to medical malpractice in the United States. Clin Orthop Relat Res 2009;467(2):339-47. doi:10.1007/s11999-008-0636-2
4. Balsamini D, Boniello K. Ex-New York Giant Michael Cox wins big in case against Manhattan doctor. *New York Post.* 2022.
5. Behm C. Former Philadelphia Eagles player wins $43.5M verdict in knee injury case. *Becker's Orthopedic Review.* Online; 2023.
6. Bokshan SL, Ruttiman R, Eltorai AEM, DePasse JM, Daniels AH, Owens BD. Factors Associated With Physician Loss in Anterior Cruciate Ligament Reconstruction Malpractice Lawsuits. Orthop J Sports Med 2017;5(11):2325967117738957. doi:10.1177/2325967117738957
7. Cichos KH, Ewing MA, Sheppard ED, Fuchs C, McGwin G, Jr., McMurtrie JT, et al. Trends and Risk Factors in Orthopedic Lawsuits: Analysis of a National Legal Database. Orthopedics 2019;42(2):e260-e7. doi:10.3928/01477447-20190211-01

Journal Pre-proof

Journal Pre-proof

8. Gatto JD, Park HY, Hwang R, Aderibigbe T, McClintick DJ, Shahinyan RH, et al. Analysis of Medical Malpractice Outcomes for Sports Orthopedic Procedures. Orthopedics 2022;45(1):e47-e52. doi:10.3928/01477447-20211101-06

9. Jena AB, Seabury S, Lakdawalla D, Chandra A. Malpractice risk according to physician specialty. N Engl J Med 2011;365(7):629-36. doi:10.1056/NEJMsa1012370

10. Minneapolis FRBo. Inflation Calculator; 2024. Available from: https://www.minneapolisfed.org/about-us/monetary-policy/inflation-calculator. [Accessed February 16th 2024 2024].

11. Navarro RA. Recent NFL Player Judgment Could Alter the Athlete-Team Physician Relationship. *AAOS Now.* American Academy of Orthopedic Surgeons; 2023.

12. Ortiz-Babilonia CD, Mo K, Raad M, Ficke JR, Jain A. Orthopaedic Surgeon Distribution in the United States. J Am Acad Orthop Surg 2022;30(18):e1188-e94. doi:10.5435/JAAOS-D-22-00271

13. Pachman S, Lamba A. Legal Aspects of Concussion: The Ever-Evolving Standard of Care. J Athl Train 2017;52(3):186-94. doi:10.4085/1062-6050-52.1.03

14. Panhuyzen-Goedkoop NM, Smeets JL. Legal responsibilities of physicians when making participation decisions in athletes with cardiac disorders: Do guidelines provide a solid legal footing? Br J Sports Med 2014;48(15):1193-5. doi:10.1136/bjsports-2013-093023

15. Rossiter A, Warrington GD, Comyns TM. Effects of Long-Haul Travel on Recovery and Performance in Elite Athletes: A Systematic Review. J Strength Cond Res 2022;36(11):3234-45. doi:10.1519/JSC.0000000000004021

Journal Pre-proof

Journal Pre-proof

16. Rynecki ND, Coban D, Gantz O, Gupta R, Ayyaswami V, Prabhu AV, et al. Medical Malpractice in Orthopedic Surgery: A Westlaw-Based Demographic Analysis. Orthopedics 2018;41(5):e615-e20. doi:10.3928/01477447-20180621-06

17. Sauder N, Emara AK, Rullan PJ, Molloy RM, Krebs VE, Piuzzi NS. Hip and Knee Are the Most Litigated Orthopaedic Cases: A Nationwide 5-Year Analysis of Medical Malpractice Claims. J Arthroplasty 2023;38(7 Suppl 2):S443-S9. doi:10.1016/j.arth.2022.11.021

18. Sharma A, Whitlock KG, Gage MJ, Lassiter TE, Anakwenze OA, Klifto CS. Malpractice trends in shoulder and elbow surgery. J Shoulder Elbow Surg 2021;30(9):2007-13. doi:10.1016/j.jse.2020.12.018

19. Silverman C, Appel CE, L.L.P. HB. Nuclear Verdicts: Trends, Causes, and Solutions. United States Chamber of Commerce; 2022:60.

20. Studdert DM, Mello MM, Sage WM, DesRoches CM, Peugh J, Zapert K, et al. Defensive medicine among high-risk specialist physicians in a volatile malpractice environment. JAMA 2005;293(21):2609-17. doi:10.1001/jama.293.21.2609

21. Young A, Chaudhry HJ, Pei X, Arnhart K, Dugan M, Snyder GB. A Census of Actively Licensed Physicians in the United States, 2016. Journal of Medical Regulation 2017;103(2):7-21. doi:10.30770/2572-1852-103.2.7

22. Zhang JK, Alimadadi A, ReVeal M, Del Valle AJ, Patel M, O'Malley DS, et al. Litigation involving sports-related spinal injuries: a comprehensive review of reported legal claims in the United States in the past 70 years. Spine J 2023;23(1):72-84. doi:10.1016/j.spinee.2022.08.012

Journal Pre-proof

Demographics	
Total lawsuits (n)	88
Mean Age (Years)	23.1
Minors (n)	32 [36]
Adult (n)	41 [47]
Female (n)	22 [25]
Male (n)	66 [75]
Year of lawsuit:	
2013-2023 (n)	31 [35]
2003-2013 (n)	42 [48]
1993-2002 (n)	15 [17]
Sport:	
Football(n)	26 [30]
Basketball (n)	9 [10]
Baseball(n)	4 [4.5]
Soccer (n)	5 [5.7]
Level of play:	
High school (n)	30 [34]
College (n)	14 [16]
Pro (n)	10 [11]
Defendant Specialty	
Orthopedic Surgery (n)	40 [45]
Internal medicine/Family medicine (n)	10 [11]
Sports Medicine (n)	5 [5.7]
Pediatrics (n)	5 [5.7]
Cardiology (n)	4 [4.5]
Other (n)	24 [27]
Treatment of Upper Extremity	14 [16]

Journal Pre-proof

Journal Pre-proof

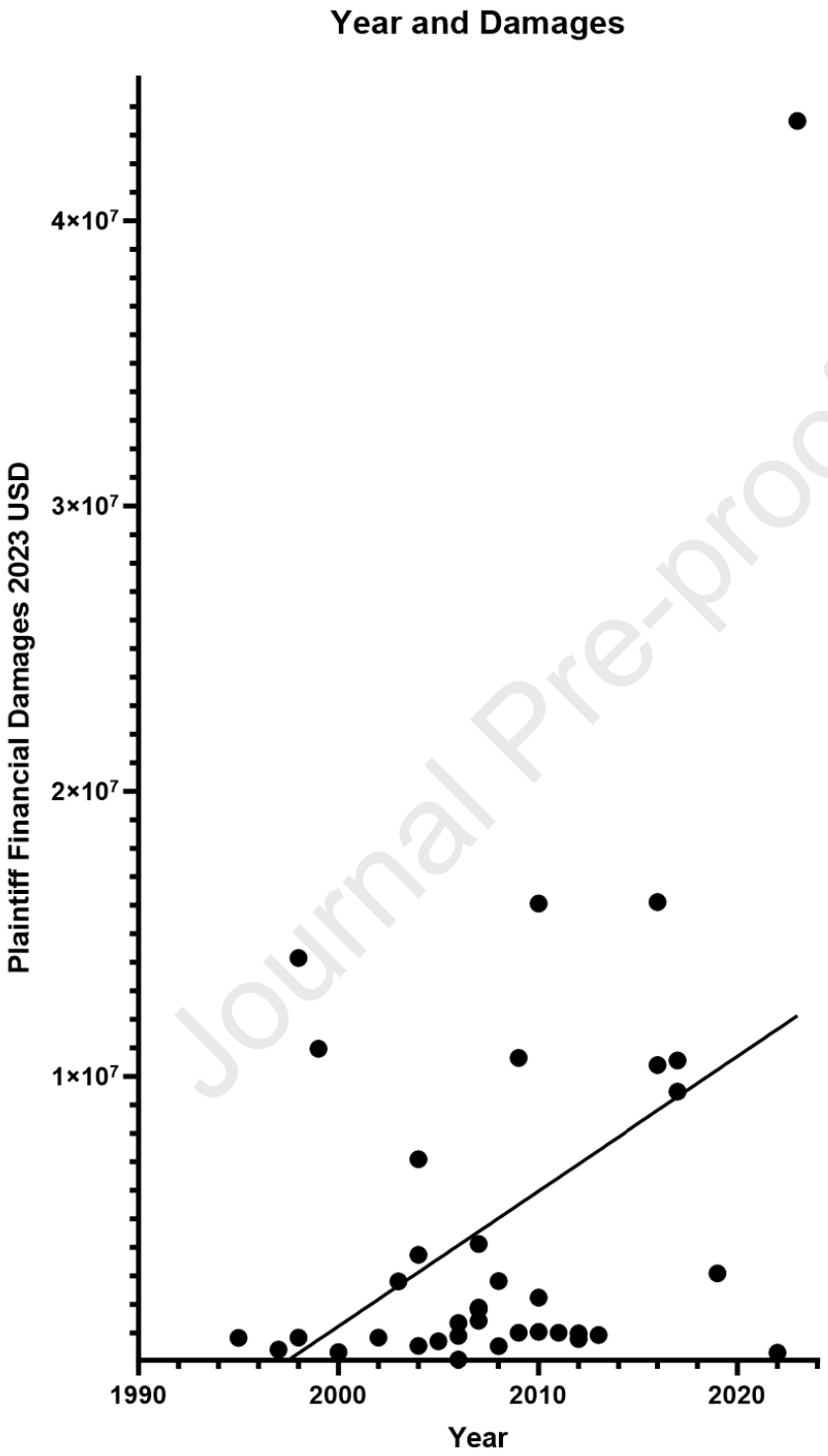
	Number of lawsuits	Decisions in favor of plaintiffs	Decisions in favor of defendants	Mean settlement (2023 USD)	Sum of total settlements (2023 USD)	Percent of sum (%)
Total cases	88	37	45	5,043,573	186,612,214	-
Level of play						
High school	30	14	14	3,724,853	52,147,945	28%
College	14	7	6	4,217,751	29,524,259	16%
Professional	10	4	6	18,340,184	77,642,895	42%
Other	34	12	19	2,274,760	27,297,115	14%
Sport						
Football	26	10	16	9,166,797	91,667,971	49%
Basketball	9	4	5	1,789,317	7,157,267	3.8%
Baseball	4	2	2	5,957,039	11,914,079	6.3%
Soccer	5	2	3	1,938,534	3,877,068	2.1%
Other	44	19	19	1,432,823	71,995,829	39%
Wrongful death	16	8	7	1,259,587	10,076,699	5.4%
Upper Extremity	14	3	11	5,314,867	15,944,602	8.5%

Journal Pre-proof

Journal Pre-proof

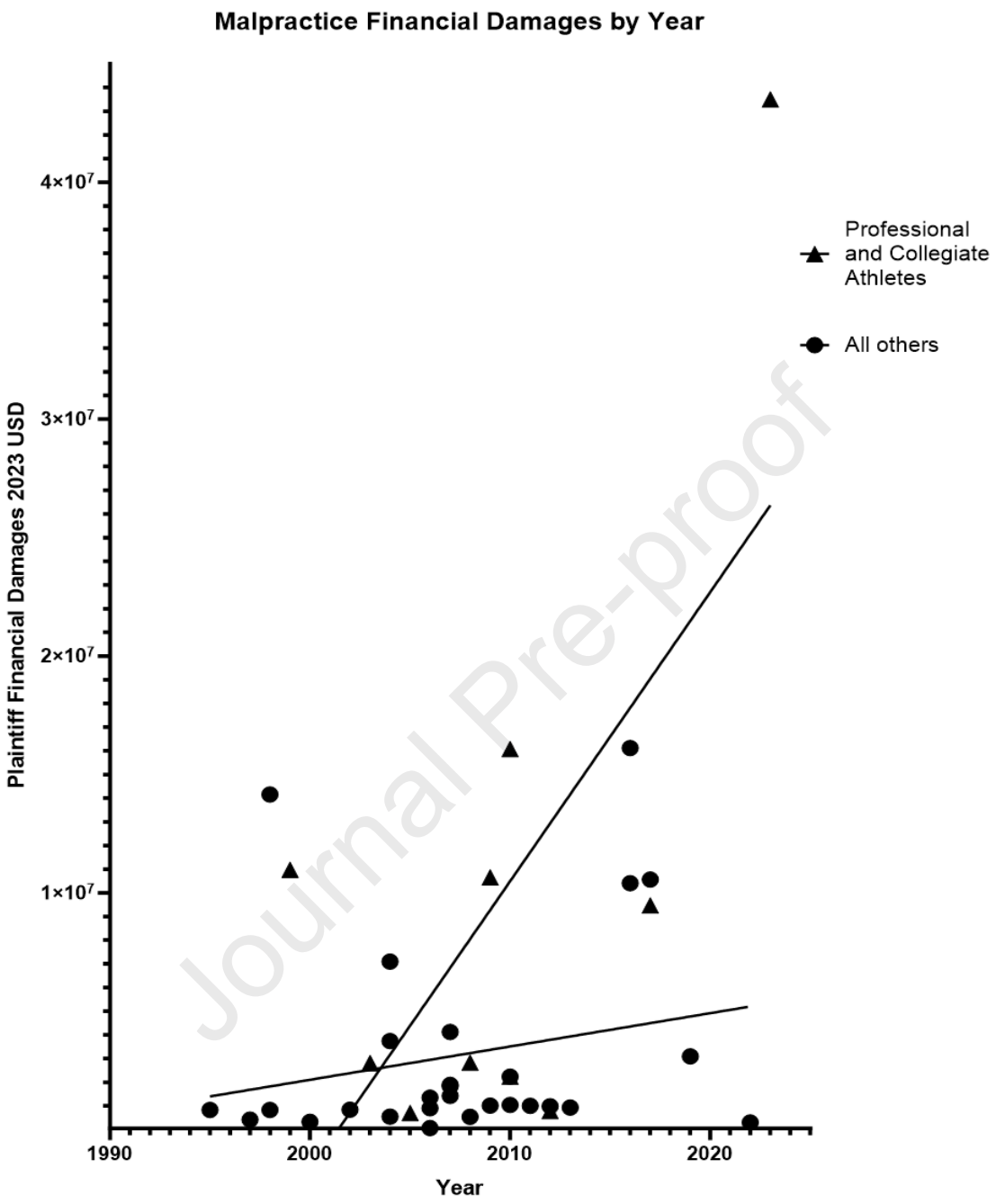
	Number of lawsuits in favor of plaintiff (n)	Median damages (2023 dollars)	p-value
Total	37	1,432,823	-
Professional and collegiate	11	2,830,448	**0.0476**
High school and all others	18	1,836,953	
Football	10	1,915,887	0.3354
All others	27	1,432,823	
Pediatric	16	3,290,044	0.0845
Adult	14	1,007,271	
Male	23	1,019,191	0.9630
Female	14	1,865,447	
Wrongful death	9	967,061	0.1275
Harm other than death	28	1,893,941	
Defendant specialty of	19	1,893,941	
orthopedics	18	1,007,271	0.3127
All other specialties			
	22	2,354,883	0.2495
Defendant surgical specialty	15	995,351	
Defendant nonsurgical specialty			

Journal Pre-proof



Journal Pre-proof

Journal Pre-proof



Journal Pre-proof